

Department of Geology and Geophysics

Subject name: Electromagnetic methods Laboratory (0-0-3) -2 Credit

Pre-requisite: None

Model experiments using electromagnetic methods. VLF and MT measurement and data interpretation, Detectability and resolution studies of EM methods. Current density cross section in VLF interpretation, Slingram configuration, focusing phenomenon in electromagnetics. Computation of MT responses over multi-layers' earth.

Text Books

1. Applied Geophysics, Telford et al, 1990
2. Geophysical Electromagnetic Theory and Methods, by MS Zhadnov, Elsevier
3. Practical Magnetotellurics, F Simpson and K Bar, Cambridge Uni. Press

Reference books

1. Potential Theory and Applied Geophysics, K.K. Roy
2. An Introduction to Geophysical Exploration, P. Keary

S.No	Topic	Number of lectures
1	EM response in free space inside a large loop rectangular transmitter	1
2	EM response in with conductor in subsurface inside a large loop rectangular transmitter	1
3	Hz, Hx and Hy responses inside a large loop with and without conductors in subsurface	3
4	Testing of Focusing phenomenon in EM prospecting	1
5	Computation of apparent current density for VLF responses	1
6	Computation of real, imaginary app. Res and phase over vertical/dipping conductor	1
7	Computation of dipole EM responses Over multi-layered earth	1
8	Computation of MT response over multi-layered earth	1
9	Interpretation of MT data over multi-layered earth	3
10	Computation of Early and layer time apparent resistivity from TEM data	2
	Total	15